

Compositional Linearization for Superposition and Hamiltonianization

Maximilien Péroux

David I. Spivak

Abstract

test

- Def of monoidal promonad. Main example is quadratic form.
- Any generally theory of quadratic forms can be translated?
- Algebra for monoidal promonads.
- Moore internalization for promondic lenses. Check if true.
- Lens for monoidal promonads.
- Quick reminder about Para.
- Monoidal product on Lin.
- Check moidal retraction/section between Lin and Crit.
- Lift to Coalg over Poly.
- Superposition and physics. Lots of examples.
- Some relation to Poincaré point of view.
- Any relation to linearization of system of differential equations? PDEs?

1 Introduction